

A polylogarithmic higher-order Cheeger inequality

Yunpeng Li*

Abstract

Let $\lambda_k(G)$ be the k th eigenvalue of the normalized Laplacian of a finite undirected weighted graph, and let $\rho_G(k)$ be the minimum possible maximum conductance of k disjoint nonempty vertex sets. We prove

$$\rho_G(k) \leq C[1 + \log(k+1)]^5 \sqrt{\lambda_k(G)}$$

for an absolute constant C . The construction gives exactly k sets and a bound in terms of $\lambda_k(G)$, with all boundaries and volumes measured in the original graph. It also yields k nonnegative functions with pairwise disjoint supports and Rayleigh quotients $O([1 + \log(k+1)]^{10} \lambda_k(G))$. A stronger conductance bound is given in Lap Chi Lau's book manuscript *Algorithmic Spectral Graph Theory* (2026); the present paper develops a different approach. The proof uses independent local cutoffs whose lost covariance is controlled by conditioning on the loss along a principal direction in each cell. A regularized spectral embedding bounds cutoff energy on the entire original low eigenspace, while an adaptive construction reduces the remaining coefficient dimension by at least half at each stage. A dimension argument then converts almost rank-one local covariances into exactly k scalar witnesses.

1 Introduction

Let G be a finite undirected graph on a vertex set V , with symmetric nonnegative edge weights (w_{vw}) and positive degrees $d_v = \sum_w w_{vw}$. Let $A = (w_{vw})$ be its adjacency matrix and $D = \text{diag}(d_v)$ its degree matrix. The combinatorial and normalized Laplacians are, respectively,

$$L = D - A \quad \text{and} \quad \mathcal{L}_G = D^{-1/2} L D^{-1/2}.$$

Self-loops are allowed. Connected components are defined using positive-weight nonloop edges. Write the eigenvalues of $\mathcal{L}_G$ as $0 = \lambda_1(G) \leq \dots \leq \lambda_{|V|}(G)$. For a nonempty vertex set S , define

$$\phi_G(S) = \frac{\sum_{v \in S, w \notin S} w_{vw}}{\sum_{v \in S} d_v}.$$

We use this one-sided conductance even when S has more than half the total volume. For $1 \leq k \leq |V|$, put

$$\rho_G(k) = \min_{\substack{S_1, \dots, S_k \neq \emptyset \\ S_i \cap S_j = \emptyset \ (i \neq j)}} \max_{1 \leq i \leq k} \phi_G(S_i). \quad (1.1)$$

The sets need not cover the graph. For a nonzero real function f on V , define its Rayleigh quotient by the following ratio, counting each unordered nonloop edge once:

$$\mathcal{R}_G(f) = \frac{\sum_{\{v,w\}} w_{vw} (f(v) - f(w))^2}{\sum_v d_v f(v)^2}. \quad (1.2)$$

*School of Data Science, The Chinese University of Hong Kong, Shenzhen. Email: liyunpeng@cuhk.edu.cn.

Lee, Oveis Gharan, and Trevisan [3] proved the higher-order Cheeger inequality

$$\frac{\lambda_k(G)}{2} \leq \rho_G(k) \leq O(k^2) \sqrt{\lambda_k(G)}.$$

The question whether the factor k^2 can be replaced by a polylogarithmic function appears in [3, Section 5.2] and in Bandeira's notes [1, Open Problem 3.3]. The question requires both the number of sets and the spectral index to equal k . A polylogarithmic bound was obtained by Lau in his book manuscript [2, Theorem 7.17]. Here we give another proof using covariance conditioning and adaptive localization of the low eigenspace. Our result is as follows.

Theorem 1.1. *There is an absolute constant C such that every graph G as above and every integer $1 \leq k \leq |V|$ admit k pairwise disjoint nonempty sets $S_1, \dots, S_k$ with*

$$\max_{1 \leq i \leq k} \phi_G(S_i) \leq C[1 + \log(k+1)]^5 \sqrt{\lambda_k(G)}. \quad (1.3)$$

There also exist nonzero nonnegative functions $f_1, \dots, f_k$ with disjoint supports satisfying

$$\mathcal{R}_G(f_i) \leq C[1 + \log(k+1)]^{10} \lambda_k(G) \quad (1 \leq i \leq k). \quad (1.4)$$

Related work. Earlier results obtained logarithmic dependence with a relaxation in the number of sets or the eigenvalue index. Louis, Raghavendra, Tetali, and Vempala [4] obtain a constant fraction of k disjoint sets of conductance $O(\sqrt{\lambda_k \log k})$. Lee, Oveis Gharan, and Trevisan [3] also prove $\rho_G(k) \leq O(\sqrt{\lambda_{2k} \log k})$ when $2k \leq |V|$. Lau [2, Theorem 7.17] obtains $\rho_G(k) \leq O((\log k)^{3/2}) \sqrt{\lambda_k(G)}$ for $k \geq 2$. Here we present a weaker bound through a different approach, using isotropic spectral embeddings and scalar sweep cuts, as in [3], and the independent-matrix Laplace inequality of Tropp [5]. The latter controls a sum of small residual covariance matrices after conditioning separately on each cell's loss along a principal covariance direction.

Proof outline. We work with the first k orthonormal eigenfunctions. Section 3 controls the sum of local covariance losses by conditioning separately on the loss along a largest-eigenvalue direction of each cell covariance and applying matrix concentration to the remaining directions. Section 4 constructs a regularized embedding whose weighted gradient is controlled on the entire original k -dimensional eigenspace. In Section 5, a Gaussian projection and a translated grid yield cell covariances whose eigenvalues other than the largest sum to at most $\delta = 1/[32N_* \log(2k)]$, where $N_* = 1 + \lceil \log_2 k \rceil$. Independent local thresholds give disjoint cutoff functions and a remainder cutoff, with covariance loss at most $1/(8N_*)$, remainder covariance trace at most one eighth of the current dimension, and Lipschitz constant $O([1 + \log(k+1)]^{7/2})$. Section 6 projects onto eigenspaces of the remainder covariance for eigenvalues above $1/4$, reducing the coefficient dimension by at least half per stage. After at most N_* stages, the resulting fields have disjoint supports, total covariance at least $(5/8)I_k$, and total energy at most $O([1 + \log(k+1)]^{10} \lambda_k) I_k$. Section 7 then shows that the local eigenvalue counts up to a constant multiple of this energy bound, each capped at two, sum to at least k . One or two nonnegative functions with disjoint supports are extracted from each selected support, and scalar sweeps give the required k sets.

2 Notation and preliminaries

We express the graph Laplacian and energy using the random walk on G . For $S \subseteq V$, write $\text{vol}(S) = \sum_{v \in S} d_v$. The probability of moving from v to w and the stationary probability measure

are, respectively,

$$P(v, w) = \frac{w_{vw}}{d_v} \quad \text{and} \quad \pi(v) = \frac{d_v}{\text{vol}(V)}.$$

Symmetry of the edge weights gives $\pi(v)P(v, w) = \pi(w)P(w, v)$. This identity is called reversibility: at stationarity, each edge is traversed with the same probability in either direction. It implies that $\Delta = I - P = D^{-1}L$ is self-adjoint in $L^2(\pi)$. This operator is similar to $\mathcal{L}_G$, so it has the same eigenvalues. For the symmetric edge weights $c_{vw} = \pi(v)P(v, w) = w_{vw}/\text{vol}(V)$, all sums indexed by (v, w) are over ordered pairs. Define the bilinear energy form by

$$\mathcal{E}(f, g) = \langle f, \Delta g \rangle_\pi = \frac{1}{2} \sum_{v, w} c_{vw} (f(v) - f(w))(g(v) - g(w)). \quad (2.1)$$

Here $\langle f, g \rangle_\pi = \sum_v \pi(v) f(v) g(v)$ and $\|f\|_\pi^2 = \langle f, f \rangle_\pi$. We abbreviate $\mathcal{E}(f, f)$ to $\mathcal{E}(f)$. For a vertex set S , write $\pi(S) = \sum_{v \in S} \pi(v)$ and $c(S, S^c) = \sum_{v \in S, w \notin S} c_{vw}$. For $S \neq \emptyset$, $\phi_G(S) = c(S, S^c)/\pi(S)$; for $f \neq 0$, $\mathcal{R}_G(f) = \mathcal{E}(f)/\|f\|_\pi^2$. For vector-valued functions, energy is the sum of coordinate energies. A function on $B \subseteq V$ is extended by zero. Thus an edge $\{v, w\}$ with $v \in B$ and $w \notin B$ contributes $c_{vw} f(v)^2$ to $\mathcal{E}(f)$, or $w_{vw} f(v)^2$ before division by $\text{vol}(V)$. The corresponding compression of Δ to $L^2(\pi|_B)$ is denoted Δ_B^D .

A field F with d columns is a linear map from $\mathbb{R}^d$ into $L^2(\pi)$, with row $F(v)$ and support $\text{supp } F = \{v : F(v) \neq 0\}$. Coefficient vectors are columns, so $(Fx)(v) = F(v)x$. Field adjoints use the π inner product; matrix adjoints in coefficient space use the Euclidean inner product. A scalar function between fields denotes multiplication on the vertex space, so $F^* f F = \sum_v \pi(v) f(v) F(v)^* F(v)$. Unless a subscript is given, vector norms are Euclidean and matrix norms are operator norms. The Frobenius norm is denoted $\|\cdot\|_F$. For symmetric matrices, $\Sigma_1 \preceq \Sigma_2$ means that $\Sigma_2 - \Sigma_1$ is positive semidefinite. The identity I acts on the space determined by context; I_d acts on $\mathbb{R}^d$. The letter C denotes an absolute positive constant that may increase between estimates. We abbreviate $\lambda_i(G)$ to λ_i when the graph is fixed. All logarithms are natural except for $\log_2$.

Fix an integer $2 \leq d \leq |V|$ with $\nu = \lambda_d(G) > 0$. Let the columns of F be $L^2(\pi)$ -orthonormal eigenfunctions of Δ for $\lambda_1, \dots, \lambda_d$, and set $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. Thus

$$F^* F = I_d, \quad \Delta F = F \Lambda, \quad 0 \preceq \Lambda \preceq \nu I_d. \quad (2.2)$$

For every coefficient vector $x \in \mathbb{R}^d$, it follows that $\|Fx\|_\pi = \|x\|$, $\mathcal{E}(Fx) \leq \nu \|x\|^2$, and $\|\Delta Fx\|_\pi \leq \nu \|x\|$. The case $\lambda_d = 0$ will be handled at the end.

For a positive semidefinite matrix Σ , define

$$\text{tail}_1(\Sigma) = \text{tr } \Sigma - \lambda_{\max}(\Sigma) = \min_{\|u\|=1} \text{tr}[(I - uu^*)\Sigma]. \quad (2.3)$$

This is its rank-one trace tail. The variational formula shows that $0 \preceq \Sigma' \preceq \Sigma$ implies $\text{tail}_1(\Sigma') \leq \text{tail}_1(\Sigma)$.

3 Selecting local losses

The next lemma chooses one local loss matrix for each cell while controlling their sum in every coefficient direction. Its hypothesis bounds the trace outside a largest-eigenvalue direction of each cell covariance. No simultaneous diagonalization is assumed.

Lemma 3.1. *Let $\ell \geq 1$ and $\delta, \alpha > 0$. Let (Σ_i) be a finite family of positive semidefinite $\ell \times \ell$ matrices such that*

$$\sum_i \Sigma_i \preceq I_\ell, \quad \text{tail}_1(\Sigma_i) \leq \delta.$$

Suppose independent random matrices (M_i) satisfy

$$0 \preceq M_i \preceq \Sigma_i, \quad \mathbb{E} M_i \preceq \alpha \Sigma_i.$$

There is a realization for which

$$\left\| \sum_i M_i \right\| \leq 4e\alpha + 2\delta \log(2\ell). \quad (3.1)$$

Proof. If $\Sigma_i = 0$, then $M_i = 0$, so omit these indices. For each remaining i , choose a unit eigenvector u_i corresponding to $\lambda_{\max}(\Sigma_i)$. Define

$$\Pi_i^{\text{top}} = u_i u_i^*, \quad \Sigma_i^\perp = \Sigma_i - \Pi_i^{\text{top}} \Sigma_i \Pi_i^{\text{top}}.$$

The projection Π_i^{top} commutes with its own covariance Σ_i . Hence $\Sigma_i^\perp$ is positive semidefinite and $\text{tr} \Sigma_i^\perp = \text{tail}_1(\Sigma_i) \leq \delta$, which implies $\Sigma_i^\perp \preceq \delta I_\ell$. Both $\Sigma_i^\perp$ and $\Pi_i^{\text{top}} \Sigma_i \Pi_i^{\text{top}}$ are bounded above by Σ_i , so

$$\sum_i \Sigma_i^\perp \preceq I_\ell, \quad \sum_i \Pi_i^{\text{top}} \Sigma_i \Pi_i^{\text{top}} \preceq I_\ell.$$

For each i , let $\mathcal{A}_i$ be the event

$$u_i^* M_i u_i \leq 2\alpha \lambda_{\max}(\Sigma_i).$$

Since $\mathbb{E}(u_i^* M_i u_i) \leq \alpha \lambda_{\max}(\Sigma_i)$, Markov's inequality gives $\mathbb{P}(\mathcal{A}_i) \geq 1/2$. Sample each M_i from its law conditioned on $\mathcal{A}_i$, independently across i . For every outcome of this product law,

$$\Pi_i^{\text{top}} M_i \Pi_i^{\text{top}} = (u_i^* M_i u_i) \Pi_i^{\text{top}} \preceq 2\alpha \Pi_i^{\text{top}} \Sigma_i \Pi_i^{\text{top}}.$$

Summing over i gives

$$\sum_i \Pi_i^{\text{top}} M_i \Pi_i^{\text{top}} \preceq 2\alpha I_\ell. \quad (3.2)$$

Define the random residual matrices by $M_i^\perp = (I_\ell - \Pi_i^{\text{top}}) M_i (I_\ell - \Pi_i^{\text{top}})$. They satisfy

$$0 \preceq M_i^\perp \preceq \Sigma_i^\perp \preceq \delta I_\ell.$$

Their conditional expectations satisfy the explicit bound

$$\mathbb{E}_{\text{cond}} M_i^\perp = \frac{\mathbb{E}(M_i^\perp \mathbf{1}_{\mathcal{A}_i})}{\mathbb{P}(\mathcal{A}_i)} \preceq 2\mathbb{E} M_i^\perp \preceq 2\alpha(I_\ell - \Pi_i^{\text{top}}) \Sigma_i (I_\ell - \Pi_i^{\text{top}}) = 2\alpha \Sigma_i^\perp.$$

In particular, $\left\| \sum_i \mathbb{E}_{\text{cond}} M_i^\perp \right\| \leq 2\alpha$.

We use the following consequence of the independent-matrix Laplace inequality [5, Theorem 3.6 and Corollary 3.7]: if the matrices $M_i^\perp$ are independent, satisfy $0 \preceq M_i^\perp \preceq \delta I_\ell$, and obey $\left\| \sum_i \mathbb{E} M_i^\perp \right\| \leq \alpha_*$, then, for every real t ,

$$\mathbb{P} \left\{ \lambda_{\max} \left(\sum_i M_i^\perp \right) \geq t \right\} \leq \ell \exp \left(\frac{-t + (e-1)\alpha_*}{\delta} \right). \quad (3.3)$$

Indeed, for every $v > 0$, the scalar chord inequality on $[0, \delta]$, applied to the eigenvalues of $M_i^\perp$, gives

$$\mathbb{E} e^{v M_i^\perp} \preceq I + \frac{e^{v\delta} - 1}{\delta} \mathbb{E} M_i^\perp \preceq \exp \left(\frac{e^{v\delta} - 1}{\delta} \mathbb{E} M_i^\perp \right).$$

The matrix logarithm is order preserving on positive definite matrices. At $v = 1/\delta$, summing the logarithmic bounds gives

$$\sum_i \log \mathbb{E} e^{M_i^\perp/\delta} \preceq \frac{e - 1}{\delta} \sum_i \mathbb{E} M_i^\perp \preceq \frac{(e - 1)\alpha_*}{\delta} I_\ell.$$

Every eigenvalue of this sum is therefore at most $(e - 1)\alpha_*/\delta$. The cited Laplace inequality yields

$$\mathbb{P}\{\lambda_{\max}(\sum_i M_i^\perp) \geq t\} \leq e^{-t/\delta} \text{tr} \exp \left(\sum_i \log \mathbb{E} e^{M_i^\perp/\delta} \right) \leq \ell \exp \left(\frac{-t + (e - 1)\alpha_*}{\delta} \right),$$

which proves (3.3). Apply it under the product conditional law with $\alpha_* = 2\alpha$ and $t = 2(e - 1)\alpha + \delta \log(2\ell)$. With probability at least $1/2$,

$$\left\| \sum_i M_i^\perp \right\| \leq 2(e - 1)\alpha + \delta \log(2\ell). \quad (3.4)$$

Fix an outcome satisfying (3.4). For every $x \in \mathbb{R}^\ell$, decompose $M_i^{1/2}x = M_i^{1/2}\Pi_i^{\text{top}}x + M_i^{1/2}(I_\ell - \Pi_i^{\text{top}})x$. The triangle inequality in the direct sum over i gives

$$\left(\sum_i x^* M_i x \right)^{1/2} \leq \left(\sum_i x^* \Pi_i^{\text{top}} M_i \Pi_i^{\text{top}} x \right)^{1/2} + \left(\sum_i x^* (I - \Pi_i^{\text{top}}) M_i (I - \Pi_i^{\text{top}}) x \right)^{1/2}.$$

Apply (3.2) and (3.4), take the supremum over $\|x\| = 1$, and use $(\sqrt{a} + \sqrt{b})^2 \leq 2a + 2b$ for $a, b \geq 0$ to obtain

$$\left\| \sum_i M_i \right\| \leq \left(\sqrt{2\alpha} + \sqrt{2(e - 1)\alpha + \delta \log(2\ell)} \right)^2 \leq 4e\alpha + 2\delta \log(2\ell). \quad \square$$

Remark 3.2. Independence is used only under the product of the separately conditioned local laws, to show that (3.4) has positive probability. The chosen deterministic family requires no further independence property. The bound does not depend on the number of cells.

4 Regularized embeddings and localization energy

The geometric construction uses a projection of the low eigenspace, while the resulting energy estimate must hold on the entire original coefficient space. We obtain this estimate by regularizing the projected field and using a weighted ground-state transform. Throughout this section, fix $\nu > 0$ and a real field F with d columns such that

$$F^*F = I_d, \quad \Delta F = F\Lambda, \quad 0 \preceq \Lambda \preceq \nu I_d. \quad (4.1)$$

The adjoint of a vertex field is taken in $L^2(\pi)$. All edge sums are over ordered pairs, with $c_{vw} = \pi(v)P(v, w) = c_{wv}$.

Lemma 4.1 (Regularized envelope). *Let $H \in \mathbb{R}^{d \times r}$, where $r \geq 1$, and put $a = 2\nu$. Regularize HH^* by the convergent series*

$$\Theta = \sum_{m=0}^{\infty} (\Lambda/a)^m HH^* (\Lambda/a)^m. \quad (4.2)$$

Let $W = F\Theta^{1/2}$, and write $\varrho(v) = \|W(v)\|$ for its pointwise norm. Define its scalar envelope by

$$h = (I + \Delta/a)^{-1} \varrho. \quad (4.3)$$

Then $h \geq 0$, and $J = \{v : h(v) > 0\}$ is a union of connected components of the graph. For $v \in J$, normalize the field by $U(v) = W(v)/h(v)$, and put $V_h(v) = \Delta h(v)/h(v)$. Define the weighted squared gradient of U on J by

$$\Gamma(U)(v) = \frac{1}{2} \sum_{w \in J} P(v, w) \frac{h(w)}{h(v)} \|U(v) - U(w)\|^2. \quad (4.4)$$

Set $U = V_h = \Gamma(U) = 0$ outside J . The normalized field and potential satisfy pointwise bounds, and the envelope has bounded total mass:

$$\begin{aligned} \|U(v)\| &\leq 2, \\ |V_h(v)| &\leq a, \\ \|h\|_{\pi}^2 &\leq 2\|H\|_{\mathbb{F}}^2. \end{aligned} \quad (4.5)$$

Moreover, every coefficient vector $x \in \mathbb{R}^d$ satisfies

$$\sum_v \pi(v) |(Fx)(v)|^2 \Gamma(U)(v) \leq 80\nu \|x\|^2. \quad (4.6)$$

There is also a contraction $Q : \mathbb{R}^r \rightarrow \mathbb{R}^d$ such that $H = \Theta^{1/2}Q$. Consequently the field $U_0 = FH/h$ on J , extended by zero outside J , satisfies $U_0 = UQ$.

Proof. Since $\|\Lambda/a\| \leq 1/2$, the series defining Θ converges in operator norm. Its summands are positive semidefinite, so $HH^* \preceq \Theta$. Shifting the index gives

$$\Lambda\Theta\Lambda = a^2 \sum_{m=1}^{\infty} (\Lambda/a)^m HH^* (\Lambda/a)^m = a^2(\Theta - HH^*) \preceq a^2\Theta. \quad (4.7)$$

Taking traces and using the same bound on Λ/a yields

$$\text{tr } \Theta = \sum_{m \geq 0} \|(\Lambda/a)^m H\|_{\mathbb{F}}^2 \leq \sum_{m \geq 0} 4^{-m} \|H\|_{\mathbb{F}}^2 = \frac{4}{3} \|H\|_{\mathbb{F}}^2. \quad (4.8)$$

These conclusions allow Θ to be singular.

By (4.1) and (4.7), each row satisfies

$$\|\Delta W(v)\|^2 = F(v)\Lambda\Theta\Lambda F(v)^* \leq a^2 F(v)\Theta F(v)^* = a^2 \varrho(v)^2. \quad (4.9)$$

This implies $\Delta\varrho \leq a\varrho$ pointwise. Indeed, at a vertex with $\varrho(v) > 0$, the vector $W(v)/\varrho(v)$ has norm one. Therefore

$$\begin{aligned}\Delta\varrho(v) &= \varrho(v) - \sum_w P(v, w) \|W(w)\| \\ &\leq \left\langle \frac{W(v)}{\varrho(v)}, W(v) - \sum_w P(v, w) W(w) \right\rangle \leq \|\Delta W(v)\| \leq a\varrho(v).\end{aligned}\tag{4.10}$$

At a vertex with $\varrho(v) = 0$, the same inequality follows from $\Delta\varrho(v) = -P\varrho(v) \leq 0$.

The resolvent is positivity preserving, as is seen from the entrywise nonnegative series

$$(I + \Delta/a)^{-1} = \frac{a}{1+a} \sum_{m=0}^{\infty} \frac{P^m}{(1+a)^m}.\tag{4.11}$$

Applying it to $(I + \Delta/a)\varrho \leq 2\varrho$ yields $\varrho \leq 2h$, while the definition of h gives $\Delta h = a(\varrho - h)$. On every connected component where ϱ is not identically zero, the series (4.11) makes h strictly positive at every vertex. On all other components $\varrho = h = 0$. Thus J is a union of components. For $v \in J$, the inequality $\varrho \leq 2h$ gives $\|U(v)\| = \varrho(v)/h(v) \leq 2$. The potential satisfies

$$V_h(v) = a \left(\frac{\varrho(v)}{h(v)} - 1 \right) \in [-a, a].\tag{4.12}$$

Since Δ is nonnegative and self-adjoint in $L^2(\pi)$, its resolvent is a contraction in that space. Hence

$$\|h\|_{\pi}^2 \leq \|\varrho\|_{\pi}^2 = \text{tr}(F^* F \Theta) = \text{tr} \Theta \leq 2\|H\|_{\mathbb{F}}^2.\tag{4.13}$$

This proves (4.5).

We next prove (4.6). If J is empty, the assertion is immediate. Otherwise, all expressions involving division by h below are restricted to J . Give J the vertex measure $\mu(v) = \pi(v)h(v)^2$. The transformed conductances are

$$c_{vw}^h = c_{vw} h(v) h(w), \quad v, w \in J.\tag{4.14}$$

The weighted Laplacian with vertex measure μ and conductances c^h acts on a scalar function $g : J \rightarrow \mathbb{R}$ by

$$(\Delta_h g)(v) = \sum_{w \in J} P(v, w) \frac{h(w)}{h(v)} (g(v) - g(w)), \quad v \in J.\tag{4.15}$$

For scalar functions $g_1, g_2 : J \rightarrow \mathbb{R}$, its Dirichlet form is

$$\mathcal{E}_h(g_1, g_2) = \frac{1}{2} \sum_{v, w \in J} c_{vw}^h (g_1(v) - g_1(w))(g_2(v) - g_2(w)).\tag{4.16}$$

Write $\mathcal{E}_h(g) = \mathcal{E}_h(g, g)$. Symmetry gives $\langle g_1, \Delta_h g_2 \rangle_{\mu} = \mathcal{E}_h(g_1, g_2)$. For every $g : J \rightarrow \mathbb{R}$, extend hg by zero outside J . Since no positive-weight edges join J to its complement, expansion at each vertex of J gives

$$\frac{\Delta(hg)}{h} = \Delta_h g + V_h g.\tag{4.17}$$

The corresponding ground-state identity for the quadratic form follows by subtracting the two edge energies:

$$\begin{aligned}
\mathcal{E}(hg) - \mathcal{E}_h(g) &= \frac{1}{2} \sum_{v,w \in J} c_{vw} [h(v)(h(v) - h(w))g(v)^2 + h(w)(h(w) - h(v))g(w)^2] \\
&= \sum_{v \in J} \pi(v) h(v) g(v)^2 \Delta h(v) \\
&= \sum_{v \in J} \pi(v) V_h(v) h(v)^2 g(v)^2.
\end{aligned} \tag{4.18}$$

The identity (4.17) applies coordinatewise to U , because $W = hU$. Thus (4.9) and $V_h \geq -a$ imply the pointwise estimate

$$\begin{aligned}
\langle U(v), (\Delta_h U)(v) \rangle &= \left\langle U(v), \frac{\Delta W(v)}{h(v)} \right\rangle - V_h(v) \|U(v)\|^2 \\
&\leq (a - V_h(v)) \|U(v)\|^2 \leq 2a \|U(v)\|^2.
\end{aligned} \tag{4.19}$$

Expanding a squared difference also gives

$$\Delta_h \|U\|^2 = 2\langle U, \Delta_h U \rangle - 2\Gamma(U). \tag{4.20}$$

The inner products in (4.19)–(4.20) are Euclidean inner products at a single vertex.

Fix a real scalar function g on J , and denote the weighted gradient quantity to be estimated by

$$\mathcal{I}_g = \sum_{v \in J} \mu(v) g(v)^2 \Gamma(U)(v). \tag{4.21}$$

Multiplying (4.20) by g^2 , integrating, and using (4.19) together with $\|U(v)\| \leq 2$ gives

$$\mathcal{I}_g \leq 8a \|g\|_\mu^2 + \frac{1}{2} |\mathcal{E}_h(g^2, \|U\|^2)|. \tag{4.22}$$

We record the edge Cauchy–Schwarz calculation, including its normalization. For $v, w \in J$, the bound on U gives

$$|\|U(v)\|^2 - \|U(w)\|^2| \leq (\|U(v)\| + \|U(w)\|) \|U(v) - U(w)\| \leq 4 \|U(v) - U(w)\|. \tag{4.23}$$

It follows that

$$\begin{aligned}
|\mathcal{E}_h(g^2, \|U\|^2)| &\leq 2 \sum_{v,w \in J} c_{vw}^h |g(v) - g(w)| |g(v) + g(w)| \|U(v) - U(w)\| \\
&\leq 2(2\mathcal{E}_h(g))^{1/2} \left(\sum_{v,w \in J} c_{vw}^h (g(v) + g(w))^2 \|U(v) - U(w)\|^2 \right)^{1/2}.
\end{aligned} \tag{4.24}$$

Using $(g(v) + g(w))^2 \leq 2(g(v)^2 + g(w)^2)$ and symmetry, the remaining sum is at most

$$4 \sum_{v,w \in J} c_{vw}^h g(v)^2 \|U(v) - U(w)\|^2 = 8\mathcal{I}_g. \tag{4.25}$$

Therefore $\frac{1}{2}|\mathcal{E}_h(g^2, \|U\|^2)| \leq 4\sqrt{\mathcal{E}_h(g)\mathcal{I}_g}$. The elementary inequality $4\sqrt{\mathcal{E}_h(g)\mathcal{I}_g} \leq \mathcal{I}_g/2 + 8\mathcal{E}_h(g)$, inserted in (4.22), proves

$$\mathcal{I}_g \leq 16(\mathcal{E}_h(g) + a\|g\|_\mu^2). \quad (4.26)$$

Now fix $x \in \mathbb{R}^d$, put $f = Fx$, and take $g = f/h$ on J . Restriction to a union of connected components can only decrease the original norm and energy. Write f_J for the restriction of f to J , extended by zero. Then $\|f_J\|_\pi^2 \leq \|f\|_\pi^2 = \|x\|^2$, and

$$\mathcal{E}(f_J) \leq \mathcal{E}(f) = x^* \Lambda x \leq \nu \|x\|^2. \quad (4.27)$$

Applying (4.18) to $g = f/h$ and using $|V_h| \leq a$, we obtain

$$\mathcal{E}_h(f/h) = \mathcal{E}(f_J) - \sum_{v \in J} \pi(v) V_h(v) f(v)^2 \leq \mathcal{E}(f_J) + a\|f_J\|_\pi^2 \leq (\nu + a)\|x\|^2. \quad (4.28)$$

Also, $\|f/h\|_\mu^2 = \|f_J\|_\pi^2 \leq \|x\|^2$. Substitution into (4.26) yields

$$\sum_v \pi(v) f(v)^2 \Gamma(U)(v) \leq 16(\nu + 2a)\|x\|^2 = 80\nu\|x\|^2, \quad (4.29)$$

as required. In particular, x has not been restricted to the column space of H .

Finally, $HH^* \preceq \Theta$ implies $\text{ran } H \subseteq \text{ran } \Theta$. Let $\Theta^{\dagger/2}$ denote the positive square root of the Moore–Penrose inverse, and set $Q = \Theta^{\dagger/2}H$. The range inclusion gives $\Theta^{1/2}Q = H$, and

$$QQ^* = \Theta^{\dagger/2}HH^*\Theta^{\dagger/2} \preceq \text{proj}_{\text{ran } \Theta} \preceq I. \quad (4.30)$$

Hence Q is a contraction and $U_0 = UQ$ on J . Outside J , $W = 0$ and $FH = WQ = 0$, so the zero extensions give the same identity there. $\square$

We localize a function f by multiplying it by a scalar cutoff z_i ; the product $z_i f$ is supported where $z_i \neq 0$. The next lemma controls the total energy of these localized functions.

Lemma 4.2 (Localization identity and energy bound). *Let $z = (z_i)$ be a finite vector of real scalar functions on the vertex set. Suppose its squared norm $s_z(v) = \sum_i z_i(v)^2$ satisfies $s_z(v) \leq 1$ at every vertex, and write $D_z(v, w) = \|z(v) - z(w)\|^2$ for its squared variation between two vertices. Every real scalar function f on the vertex set satisfies*

$$\sum_i \mathcal{E}(z_i f) = \langle s_z f, \Delta f \rangle_\pi + \frac{1}{2} \sum_{v, w} c_{vw} D_z(v, w) f(v) f(w). \quad (4.31)$$

Let U be an embedding supplied by Lemma 4.1. Suppose that, for some $\kappa \geq 0$, every pair of vertices with $c_{vw} > 0$ satisfies

$$D_z(v, w) \leq \kappa^2 \|U(v) - U(w)\|^2. \quad (4.32)$$

Then, for every $x \in \mathbb{R}^d$ and $f = Fx$, the absolute value of the edge correction in (4.31) is at most $80\kappa^2\nu\|x\|^2$. In particular,

$$\sum_i \mathcal{E}(z_i Fx) \leq (1 + 80\kappa^2)\nu\|x\|^2. \quad (4.33)$$

More generally, let (U_q) be a finite collection of embeddings supplied by Lemma 4.1 for the same F, Λ, ν , with possibly different generating matrices and envelopes. For fixed numbers $\vartheta_q \geq 0$, suppose that every pair with $c_{vw} > 0$ satisfies

$$D_z(v, w) \leq \sum_q \vartheta_q \|U_q(v) - U_q(w)\|^2. \quad (4.34)$$

Then every $x \in \mathbb{R}^d$ satisfies

$$\sum_i \mathcal{E}(z_i Fx) \leq \left(1 + 80 \sum_q \vartheta_q\right) \nu \|x\|^2. \quad (4.35)$$

Proof. Fix a real scalar function f . For each ordered edge, expansion gives

$$\sum_i (z_i(v)f(v) - z_i(w)f(w))^2 = s_z(v)f(v)^2 + s_z(w)f(w)^2 - 2\langle z(v), z(w) \rangle f(v)f(w). \quad (4.36)$$

Substitute the identity

$$D_z(v, w) = s_z(v) + s_z(w) - 2\langle z(v), z(w) \rangle. \quad (4.37)$$

Multiply the resulting identity by $c_{vw}/2$ and sum over all ordered pairs. Interchange v and w in the terms containing $s_z(w)$. Since $c_{vw} = c_{wv}$, these give the same sums as the corresponding terms containing $s_z(v)$. Also, $\sum_w c_{vw} = \pi(v) \sum_w P(v, w) = \pi(v)$. Consequently,

$$\begin{aligned} \sum_i \mathcal{E}(z_i f) - \frac{1}{2} \sum_{v,w} c_{vw} D_z(v, w) f(v) f(w) \\ &= \sum_{v,w} c_{vw} s_z(v) f(v) (f(v) - f(w)) \\ &= \sum_v \pi(v) s_z(v) f(v) \left(f(v) - \sum_w P(v, w) f(w) \right) \\ &= \langle s_z f, \Delta f \rangle_\pi. \end{aligned}$$

The last equality uses $\Delta = I - P$. This proves (4.31) without requiring $s_z = 1$.

Now fix $x \in \mathbb{R}^d$ and let $f = Fx$. The first term in that identity is bounded by

$$\langle s_z f, \Delta f \rangle_\pi \leq \|s_z f\|_\pi \|\Delta f\|_\pi \leq \|Fx\|_\pi \|F\Delta x\|_\pi \leq \nu \|x\|^2. \quad (4.38)$$

This estimate uses $0 \leq s_z \leq 1$ and the norm bound on Λ ; it requires no commutation of scalar multiplication with Δ .

On a component where $h = 0$, (4.32) forces $D_z = 0$ on every edge. Thus the edge correction receives no contribution from such components. For $v, w \in J = \{h > 0\}$, the scalar inequality

$$|f(v)f(w)| \leq \frac{1}{2} \left(\frac{h(w)}{h(v)} f(v)^2 + \frac{h(v)}{h(w)} f(w)^2 \right) \quad (4.39)$$

and edge symmetry give

$$\begin{aligned} \frac{1}{2} \sum_{v,w} c_{vw} D_z(v, w) |f(v)f(w)| &\leq \frac{\kappa^2}{4} \sum_{v,w \in J} c_{vw} \|U(v) - U(w)\|^2 \left(\frac{h(w)}{h(v)} f(v)^2 + \frac{h(v)}{h(w)} f(w)^2 \right) \\ &= \kappa^2 \sum_v \pi(v) f(v)^2 \Gamma(U)(v) \\ &\leq 80\kappa^2 \nu \|x\|^2. \end{aligned} \quad (4.40)$$

The absolute value of the signed edge correction is bounded by the left side. Combining this with (4.38) proves (4.33).

Under (4.34), first bound the absolute correction by

$$\sum_q \frac{\vartheta_q}{2} \sum_{v,w} c_{vw} \|U_q(v) - U_q(w)\|^2 |f(v)f(w)|. \quad (4.41)$$

For each q , apply (4.39) with that embedding's own envelope h_q , and then use (4.6). On a zero-envelope component the corresponding squared difference is zero, so no ratio of vanishing envelope values occurs. Each summand is at most $80\vartheta_q\nu\|x\|^2$, which proves (4.35). $\square$

5 One disjoint stage

Fix the original eigenspace dimension $d \geq 2$. Set $N_* = 1 + \lceil \log_2 d \rceil$, which will bound the number of stages, and $\tau = 1/8$. The transition-width parameter, covariance-tail tolerance, and covariance-loss tolerance are, respectively,

$$\sigma = \frac{1}{128eN_*}, \quad \delta = \frac{1}{32N_* \log(2d)}, \quad \varepsilon = \frac{1}{8N_*}. \quad (5.1)$$

In particular $0 < \sigma, \delta < 1/4$. Recall that $\text{tail}_1(\Sigma) = \text{tr } \Sigma - \lambda_{\max}(\Sigma)$ denotes the rank-one trace tail of a positive semidefinite matrix.

Proposition 5.1 (One disjoint stage). *Let $Z \in \mathbb{R}^{d \times \ell}$ satisfy $Z^*Z = I_\ell$, where $1 \leq \ell \leq d$, and put $Y = FZ$. There are finitely many nonnegative cutoff functions χ_i , a nonnegative remainder cutoff β , and an embedding U supplied by Lemma 4.1, such that the supports of the cutoffs are pairwise disjoint and*

$$\sum_i \chi_i^2 + \beta^2 \leq 1, \quad Y^* \left(1 - \sum_i \chi_i^2 - \beta^2 \right) Y \preceq \varepsilon I_\ell, \quad (5.2)$$

$$\text{tr}(Y^* \beta^2 Y) \leq \tau \ell, \quad \text{tail}_1(Y^* \chi_i^2 Y) \leq \delta. \quad (5.3)$$

The vector of cutoffs (χ_i, β) is κ -Lipschitz in U : for every $v, w \in V$,

$$\sum_i |\chi_i(v) - \chi_i(w)|^2 + |\beta(v) - \beta(w)|^2 \leq \kappa^2 \|U(v) - U(w)\|^2.$$

Its Lipschitz constant satisfies

$$\kappa \leq C[1 + \log(d+1)]^{7/2} \quad (5.4)$$

for an absolute constant C . These conclusions hold for every Z with orthonormal columns; its range need not be invariant under Λ .

The cutoffs take values in $[0, 1]$. Multiplication by χ_i selects a local piece of Y , while βY is the remainder used in subsequent stages. The loss of squared norm is measured separately by $1 - \sum_i \chi_i^2 - \beta^2$.

We first obtain a low-dimensional projection that preserves most weighted pairs of directions. For every vertex v , define $m_v = \pi(v)\|Y(v)\|^2$. For those vertices with $Y(v) \neq 0$, set

$$\hat{y}_v = \frac{Y(v)}{\|Y(v)\|}.$$

Rows with $Y(v) = 0$ have zero target mass and are omitted from sums involving $\hat{y}_v$. Since $Y^*Y = I_\ell$, these definitions give

$$\sum_v m_v \hat{y}_v^* \hat{y}_v = I_\ell, \quad \sum_v m_v = \ell. \quad (5.5)$$

Choose the angular threshold and the projection dimension as

$$\theta = \arcsin \sqrt{\delta/4} \quad \text{and} \quad r = \left\lceil \max \left\{ 512 \log \frac{64\ell^2}{\delta^2}, 32 \log \frac{32}{\tau} \right\} \right\rceil. \quad (5.6)$$

An ordered pair (v, w) with nonzero target rows is *far* if $|\hat{y}_v \cdot \hat{y}_w| \leq \cos \theta$, and *near* otherwise. The projective angle between two nonzero vectors is the smaller angle between the lines they span, and lies in $[0, \pi/2]$.

Lemma 5.2 (Weighted Gaussian projection). *There is an $\ell \times r$ real matrix Ω such that*

$$\sum_{\substack{(v,w) \text{ far} \\ \angle_{\text{proj}}(\hat{y}_v \Omega, \hat{y}_w \Omega) < \theta/2}} m_v m_w \leq \delta^2/2, \quad (5.7)$$

$$\sum_{\|Y(v)\Omega/\sqrt{r}\|^2 < \|Y(v)\|^2/2} m_v \leq \tau\ell/4. \quad (5.8)$$

The projected generating matrix also satisfies

$$\|Z\Omega/\sqrt{r}\|_{\text{F}}^2 \leq 4\ell. \quad (5.9)$$

Moreover, no nonzero row of Y projects to zero. In (5.7), zero projected rows may equivalently be counted as angular failures before this final condition is imposed.

Proof. Choose the entries of Ω independently from the standard normal distribution. Here χ_r^2 denotes a chi-square random variable with r degrees of freedom. Its moment-generating function $\mathbb{E}e^{t\chi_r^2} = (1 - 2t)^{-r/2}$ for $t < 1/2$, together with the exponential Markov inequality, gives, for $0 < s < 1$,

$$\mathbb{P}\{|\chi_r^2/r - 1| > s\} \leq 2e^{-rs^2/8}. \quad (5.10)$$

Indeed, optimizing the parameter separately in the two tails gives exponents $-r[s - \log(1 + s)]/2$ and $-r[-s - \log(1 - s)]/2$. Both bracketed expressions are at least $s^2/4$: integrate $t/(1 + t) \geq t/2$ for the upper tail and $t/(1 - t) \geq t$ for the lower tail, over $0 \leq t \leq s$.

For a fixed two-dimensional subspace of $\mathbb{R}^\ell$, let e_1, e_2 be an orthonormal basis. Apply (5.10), with $s = 1/8$, to the four unit vectors $e_1, e_2, (e_1 + e_2)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}$. Except on an event of probability at most $8e^{-r/512}$, all four squared norms after projection by $\Omega/\sqrt{r}$ differ from one by at most $1/8$. The Gram matrix of $e_1\Omega/\sqrt{r}, e_2\Omega/\sqrt{r}$ then has diagonal errors at most $1/8$. Its off-diagonal entry is

$$\left\langle \frac{e_1\Omega}{\sqrt{r}}, \frac{e_2\Omega}{\sqrt{r}} \right\rangle = \frac{1}{2} \left(\left\| \frac{(e_1 + e_2)\Omega}{\sqrt{2r}} \right\|^2 - \left\| \frac{(e_1 - e_2)\Omega}{\sqrt{2r}} \right\|^2 \right),$$

whose magnitude is at most $1/8$. The two Gram eigenvalues therefore lie in $[3/4, 5/4]$.

For any two vectors in this subspace, the area of the parallelogram they span after projection is at least $3/4$ times its original area, while the product of projected lengths is at most $5/4$ times

the original product. Thus the sine of their projected angle is at least $3/5$ times its original value. If the original pair is far, its projective angle is at least θ . Since $\theta < \pi/6$,

$$\frac{3}{5} \sin \theta \geq \sin(\theta/2),$$

so its projective angle after projection is at least $\theta/2$. Summing the failure probabilities with the weights in (5.5) gives

$$\begin{aligned} \mathbb{E} \sum_{\substack{(v,w) \text{ far} \\ \angle_{\text{proj}}(\hat{y}_v \Omega, \hat{y}_w \Omega) < \theta/2}} m_v m_w &\leq 8\ell^2 e^{-r/512} \leq \delta^2/8, \\ \mathbb{E} \sum_{\|Y(v)\Omega/\sqrt{r}\|^2 < \|Y(v)\|^2/2} m_v &\leq 2\ell e^{-r/32} \leq \tau\ell/16, \\ \mathbb{E} \|Z\Omega/\sqrt{r}\|_{\text{F}}^2 &= \ell. \end{aligned} \tag{5.11}$$

The second estimate uses (5.10) with $s = 1/2$; the third uses $Z^*Z = I_\ell$. When $\ell = 1$, there are no far pairs, so the first estimate is immediate.

Each expectation in (5.11) is at most one quarter of its required upper bound in (5.7)–(5.9). Markov's inequality and a union bound over the three failure events therefore give simultaneous success probability at least $1/4$. For each fixed nonzero row of Y , its image under Ω is a nondegenerate Gaussian vector and is zero with probability zero. There are finitely many such rows, so excluding all these zero images preserves the success probability. The chosen value of r has no dependence on the number of vertices. $\square$

The next observation converts the pair estimate into a statement about every raw cell, before its cutoff is chosen.

Lemma 5.3 (Covariance tail forces far pairs). *With the notation of (5.5), let B be any set of vertices with $Y(v) \neq 0$, and put $\Sigma = \sum_{v \in B} m_v \hat{y}_v^* \hat{y}_v$. If $\text{tail}_1(\Sigma) > \delta$, then*

$$\sum_{\substack{v, w \in B \\ |\hat{y}_v \cdot \hat{y}_w| \leq \cos \theta}} m_v m_w > \delta^2/2.$$

Proof. Write $m = \text{tr } \Sigma$. If $m \leq 2$, then

$$\sum_{v, w \in B} m_v m_w [1 - (\hat{y}_v \cdot \hat{y}_w)^2] = m^2 - \text{tr}(\Sigma^2) \geq m(m - \lambda_{\max} \Sigma) > m\delta,$$

because each eigenvalue of Σ is at most $\lambda_{\max}(\Sigma)$. For a near pair the factor $1 - (\hat{y}_v \cdot \hat{y}_w)^2$ is at most $\sin^2 \theta = \delta/4$. The near-pair contribution is therefore at most $(\delta/4)m^2 \leq m\delta/2$. For a far pair that factor is at most one, so the far-pair weight exceeds $m\delta/2 > \delta^2/2$, using $m \geq \text{tail}_1(\Sigma) > \delta$.

If $m > 2$, isotropy shows that the mass of the full row family in a near cap about any unit vector $\hat{y}$ is at most $1/\cos^2 \theta$:

$$\cos^2 \theta \sum_{|\hat{y} \cdot \hat{y}_v| > \cos \theta} m_v \leq \sum_v m_v (\hat{y} \cdot \hat{y}_v)^2 = 1.$$

Since $\delta < 1/4$, the cap mass is at most $4/3$. For each $v \in B$, the far rows within B thus have total mass at least $m - 4/3$. Multiply by m_v and sum over $v \in B$ to obtain far-pair weight at least $m(m - 4/3) > \delta^2/2$. $\square$

Proof of Proposition 5.1. Fix Ω from Lemma 5.2, and apply Lemma 4.1 with the generating matrix $H = Z\Omega/\sqrt{r}$. Let h and U be the envelope and embedding it supplies. Set $U_0 = FH/h$ where $h > 0$, with $U_0 = 0$ elsewhere. The lemma supplies a contraction Q , and (4.5) together with (5.9) gives

$$U_0 = UQ, \quad \|Q\| \leq 1, \quad \|h\|_\pi^2 \leq 2\|H\|_F^2 \leq 8\ell. \quad (5.12)$$

On a zero-envelope component, $FH = 0$. Because Ω annihilates no nonzero target row, $Y = 0$ on that component as well.

Raw cells and their covariance. Use the radial scale $r_0 = \sqrt{\tau}/32$. Choose the cell diameter and side length as

$$\eta = r_0 \sin(\theta/4) \quad \text{and} \quad b = \eta/\sqrt{r}, \quad (5.13)$$

respectively, and set the face-collar width to $\omega = \tau b/(48r)$. Consider a translated grid of side b in $\mathbb{R}^r$. Assign each face point to one of its adjacent cells by a fixed rule, so that every graph row belongs to a unique cell. Call a cell active if it contains a graph row $U_0(v)$ with norm greater than r_0 . Only finitely many cells are active. Each cell has diameter $\eta < r_0$. Every point in the cell containing 0 has norm at most η , so that cell is inactive. In particular, the vector of cutoffs on every zero-envelope component will be constant.

Every row in an active cell has norm greater than $r_0 - \eta$. Two vectors of such norms whose projective angle is at least $\theta/2$ have distance at least

$$2(r_0 - \eta) \sin(\theta/4) = 2[1 - \sin(\theta/4)]\eta > \eta.$$

They cannot lie in the same cell. For an original far pair in an active cell, the projective angle of the U_0 -rows is the same as that of the projected target rows: the factors $h(v)$ and $\|Y(v)\|$ are positive. Every such pair therefore appears in the sum in (5.7).

For each active cell i , let V_i be the set of vertices whose U_0 -rows belong to that cell, and define

$$\Sigma_i = Y^* \mathbf{1}_{V_i} Y.$$

Its far-pair weight is at most $\delta^2/2$, so Lemma 5.3 implies $\text{tail}_1(\Sigma_i) \leq \delta$. The vertex sets V_i are disjoint and $Y^*Y = I_\ell$. Thus

$$\text{tail}_1(\Sigma_i) \leq \delta \quad \text{for every active cell } i, \quad \sum_i \Sigma_i \preceq I_\ell. \quad (5.14)$$

This holds for every grid shift and concerns the entire raw cell.

A shift with small collar mass. For every row satisfying $\|Y(v)\Omega/\sqrt{r}\|^2 \geq \|Y(v)\|^2/2$ and $\|U_0(v)\| < 3r_0$, the identity $FH = Y\Omega/\sqrt{r} = hU_0$ gives

$$\|Y(v)\|^2 \leq 2\|Y(v)\Omega/\sqrt{r}\|^2 \leq 18r_0^2 h(v)^2.$$

The rows that fail the norm condition have total mass at most $\tau\ell/4$ by (5.8). Therefore

$$\sum_{\|U_0(v)\| < 3r_0} m_v \leq 18r_0^2 \|h\|_\pi^2 + \tau\ell/4 \leq (144/1024 + 1/4)\tau\ell < \tau\ell/2. \quad (5.15)$$

Choose the grid shift uniformly in $[0, b)^r$. A fixed point lies within 3ω of a grid face with probability at most $6r\omega/b = \tau/8$, by a union bound over coordinates. The expected target mass in these face collars is therefore at most $\tau\ell/8$. By Markov's inequality, their mass is at most $\tau\ell/2$ for a set of shifts of probability at least $3/4$. Fix such a shift. By (5.15), the union of the face collars and the origin ball has target mass at most $\tau\ell$.

Disjoint cutoffs and their loss. Independently for each active cell choose t_i^∂ uniformly in $[\omega, 2\omega]$ and t_i^{rad} uniformly in $[r_0, 2r_0]$. For $y \in \mathbb{R}^r$ in this cell, let $d_\partial(y)$ be its distance to the cell boundary. Writing $t_+ = \max\{t, 0\}$, define

$$q_i(y) = \min \left\{ 1, \left(\frac{d_\partial(y) - t_i^\partial}{\sigma\omega} \right)_+, \left(\frac{\|y\| - t_i^{\text{rad}}}{\sigma r_0} \right)_+ \right\}, \quad (5.16)$$

$$\chi_i(y) = (2q_i(y) - 1)_+, \quad \beta(y) = (1 - 2q_i(y))_+. \quad (5.17)$$

All other cutoff functions vanish in this cell. On inactive cells put every $\chi_i = 0$ and $\beta = 1$. At most one coordinate of the vector of cutoffs is positive. Define its squared-norm deficit by

$$\zeta(y) = 1 - \sum_i \chi_i(y)^2 - \beta(y)^2.$$

On an active cell, the definitions in (5.17) give

$$\zeta(y) = 1 - (2q_i(y) - 1)^2 \in [0, 1], \quad (5.18)$$

and on inactive cells $\zeta(y) = 0$. We use the same symbols for the pullbacks to V under U_0 : evaluating at $y = U_0(v)$, we write $\chi_i(v), \beta(v), \zeta(v)$ for these values.

For a fixed row in cell i , a positive deficit requires $0 < q_i < 1$. At least one ramp in (5.16) must then lie strictly between zero and one. The boundary ramp does so only if

$$d_\partial(y) - \sigma\omega < t_i^\partial < d_\partial(y),$$

an interval of length $\sigma\omega$, intersected with a sampling interval of length ω , so its probability is at most σ . The radial condition similarly restricts t_i^{rad} to an interval of length σr_0 , intersected with its sampling interval of length r_0 . Since $0 \leq \zeta(v) \leq 1$, a union bound gives $\mathbb{E}\zeta(v) \leq 2\sigma$ for every v . Define the matrices

$$M_i = Y^* \mathbf{1}_{V_i} \zeta Y.$$

Each uses only the two random thresholds of its own cell, so these matrices are independent. They satisfy $0 \preceq M_i \preceq \Sigma_i$ and $\mathbb{E}M_i \preceq 2\sigma\Sigma_i$. Apply Lemma 3.1 with $\alpha = 2\sigma$ to choose all thresholds so that

$$Y^* \zeta Y = \sum_i M_i \preceq [8e\sigma + 2\delta \log(2\ell)] I_\ell \preceq \frac{1}{8N_*} I_\ell = \varepsilon I_\ell. \quad (5.19)$$

The last bound uses $8e\sigma = 1/(16N_*)$ and $2\delta \log(2\ell) \leq 1/(16N_*)$, since $\ell \leq d$. This proves (5.2).

Remainder, tails, and continuity. The remaining assertions hold for every allowed threshold choice, so they also hold for the selected one. If an active row has $\beta > 0$, then $q_i < 1/2$. At least one ramp is less than $1/2$, placing the row either within $2\omega + \sigma\omega/2 < 3\omega$ of a face or within $2r_0 + \sigma r_0/2 < 3r_0$ of the origin. Inactive graph rows have norm at most r_0 . All rows with $\beta > 0$ therefore lie in the union of the face collars and the origin ball. Since $\beta^2 \leq 1$, this gives $\text{tr}(Y^* \beta^2 Y) \leq \tau\ell$. For each i , the cutoff χ_i is supported on V_i and satisfies $0 \leq \chi_i^2 \leq 1$, hence $0 \preceq Y^* \chi_i^2 Y \preceq \Sigma_i$. Monotonicity of the rank-one trace tail and (5.14) prove the other assertion in (5.3).

Both distance to the boundary and Euclidean norm are 1-Lipschitz. Taking a positive part or a minimum preserves a common Lipschitz bound. Since $\omega < r_0$, the function q_i in (5.16) is therefore $1/(\sigma\omega)$ -Lipschitz inside its cell. The map

$$q \longmapsto ((2q - 1)_+, (1 - 2q)_+), \quad 0 \leq q \leq 1,$$

is 2-Lipschitz. On the same side of $1/2$ this follows directly from its slope. For inputs $q \geq 1/2 \geq q'$, the distance between their images is at most the sum of the two nonzero coordinates, $(2q - 1) + (1 - 2q') = 2(q - q')$. At every face, all $\chi_i = 0$ and $\beta = 1$, independent of the local parameters. Subdivide a line segment at its grid-face crossings and sum the within-cell bounds. This proves that the entire vector is globally $2/(\sigma\omega)$ -Lipschitz in U_0 ; a segment contained in a face has a constant cutoff vector. By (5.12) it has the same Lipschitz bound in U .

Finally, concavity of sine gives $\sin(\theta/4) \geq \sin \theta/4 = \sqrt{\delta}/8$. From (5.13),

$$\kappa = \frac{2}{\sigma\omega} \leq \frac{24576 r^{3/2}}{\sigma\tau^{3/2}\sqrt{\delta}}. \quad (5.20)$$

Equations (5.1) and (5.6) give $r = O(1 + \log(d + 1))$, $\sigma^{-1} = O(1 + \log(d + 1))$, and $\delta^{-1/2} = O(1 + \log(d + 1))$. Substituting in (5.20) proves (5.4). $\square$

6 Adaptive localization of the low eigenspace

We now apply Proposition 5.1 repeatedly, with the parameters fixed in (5.1).

Proposition 6.1. *There is an absolute constant C such that the eigenfield (2.2) admits finitely many fields X_i with pairwise disjoint supports and*

$$\sum_i X_i^* X_i \succeq \frac{5}{8} I_d \quad \text{and} \quad \sum_i X_i^* \Delta X_i \preceq C[1 + \log(d + 1)]^{10} \nu I_d. \quad (6.1)$$

Each covariance also satisfies $\text{tail}_1(X_i^* X_i) \leq \delta$, where $\delta = 1/(32N_* \log(2d)) < 1/4$.

Proof. Start with the coefficient projection $\Pi_0 = I_d$ and the accumulated remainder cutoff $\bar{\beta}_0 = 1$. At stage j , while $\Pi_{j-1} \neq 0$, set $\ell_j = \text{rank } \Pi_{j-1}$ and choose $Z_j \in \mathbb{R}^{d \times \ell_j}$ whose columns form an orthonormal basis of $\text{ran } \Pi_{j-1}$. Apply Proposition 5.1 to $Y_j = FZ_j$. The proposition applies to every such basis after the earlier choices have been fixed; independence between stages is unnecessary. For the resulting cutoff functions $\chi_{j,i}$ and remainder cutoff β_j , define the accumulated remainder cutoff and the fields selected at stage j by

$$\bar{\beta}_j = \bar{\beta}_{j-1} \beta_j \quad \text{and} \quad X_{j,i} = \bar{\beta}_{j-1} \chi_{j,i} F \Pi_{j-1},$$

respectively. The covariance of the remainder before the next projection is

$$\tilde{R}_j = \Pi_{j-1} F^* \bar{\beta}_j^2 F \Pi_{j-1}.$$

Let $\Pi_j \preceq \Pi_{j-1}$ be its spectral projection onto the eigenspaces with eigenvalues greater than $1/4$. Define the projected covariance and the discarded part by

$$R_j = \Pi_j \tilde{R}_j \Pi_j = \Pi_j F^* \bar{\beta}_j^2 F \Pi_j \quad \text{and} \quad R_j^{\text{disc}} = \tilde{R}_j - R_j.$$

For the squared-norm deficit $\zeta_j = 1 - \sum_i \chi_{j,i}^2 - \beta_j^2$, the lost covariance is

$$K_j = \Pi_{j-1} F^* \bar{\beta}_{j-1}^2 \zeta_j F \Pi_{j-1}.$$

Set $R_0 = I_d$. Multiply the identity $1 = \sum_i \chi_{j,i}^2 + \beta_j^2 + \zeta_j$ by $\bar{\beta}_{j-1}^2$, apply $F^*(\cdot)F$, and compress to $\text{ran } \Pi_{j-1}$. Since $\tilde{R}_j = R_j^{\text{disc}} + R_j$, this gives

$$R_{j-1} = \sum_i X_{j,i}^* X_{j,i} + K_j + R_j^{\text{disc}} + R_j. \quad (6.2)$$

Since $Z_j Z_j^* = \Pi_{j-1}$, the trace bound for the stage remainder and $\bar{\beta}_{j-1}^2 \leq 1$ give

$$\text{tr } \tilde{R}_j = \text{tr}(Y_j^* \bar{\beta}_{j-1}^2 \beta_j^2 Y_j) \leq \text{tr}(Y_j^* \beta_j^2 Y_j) \leq \ell_j/8.$$

The eigenvalues of $\tilde{R}_j$ on $\text{ran } \Pi_j$ exceed $1/4$, so $(\text{rank } \Pi_j)/4 < \ell_j/8$ whenever $\text{rank } \Pi_j > 0$. Thus the rank decreases by at least half at every stage. Since $d/2^{N_*} < 1$, the construction reaches $\Pi_N = 0$ for some $N \leq N_*$.

Coverage and disjoint supports. The spectral projection Π_j splits $\tilde{R}_j$ into orthogonal blocks, with eigenvalues at most $1/4$ on the complementary block. Thus

$$0 \preceq R_j^{\text{disc}} \preceq \frac{1}{4}(\Pi_{j-1} - \Pi_j).$$

The differences $\Pi_{j-1} - \Pi_j$ are mutually orthogonal projections, and their sum is $\Pi_0 - \Pi_N = I_d$. Hence $\sum_j R_j^{\text{disc}} \preceq I_d/4$. Since $\zeta_j \geq 0$ and $\bar{\beta}_{j-1}^2 \leq 1$, the stage loss bound gives

$$0 \preceq K_j \preceq \Pi_{j-1} F^* \zeta_j F \Pi_{j-1} = Z_j (Y_j^* \zeta_j Y_j) Z_j^* \preceq \varepsilon \Pi_{j-1}.$$

Summing over the at most N_* stages yields $\sum_j K_j \preceq N_* \varepsilon I_d = I_d/8$. Telescoping (6.2) therefore yields

$$\sum_{j,i} X_{j,i}^* X_{j,i} \succeq \frac{5}{8} I_d. \quad (6.3)$$

Within a stage the cutoff functions have disjoint supports. Where $\chi_{j,i} > 0$, we have $\beta_j = 0$, so all later accumulated remainder cutoffs vanish there. The fields $X_{j,i}$ therefore have pairwise disjoint supports across all stages.

The covariance-tail transfer is explicit:

$$X_{j,i}^* X_{j,i} = Z_j [Y_j^* \bar{\beta}_{j-1}^2 \chi_{j,i}^2 Y_j] Z_j^*.$$

The bracketed covariance is bounded above by $Y_j^* \chi_{j,i}^2 Y_j$. Its trace tail is at most δ by (5.3) and monotonicity in (2.3). The outer isometric embedding adds only zero eigenvalues and hence preserves this bound.

Energy of the scalar cutoffs. Write

$$\psi_{j,i} = \chi_{j,i} \prod_{q < j} \beta_q.$$

Set $\chi_j = (\chi_{j,i})_i$ and $z_j = (\chi_j, \beta_j)$. Temporarily include the final remainder cutoff $\prod_{q \leq N} \beta_q$ as one additional coordinate of ψ . Define the weighted suffix vectors recursively by

$$\xi_{N+1} = 1, \quad \xi_q = (\chi_q, \beta_q \xi_{q+1}) \quad (q = N, \dots, 1).$$

Then $\psi = \xi_1$, and $\|\xi_q\| \leq 1$ follows inductively from $\|z_q\| \leq 1$. This holds for any tuple of stage vectors in their unit balls, including tuples mixing values at different vertices.

In the algebraic expression for ψ , vary only z_j and hold the other stage vectors fixed. The affected coordinates are $(\prod_{q < j} \beta_q)(\chi_j, \beta_j \xi_{j+1})$. The prefix product has absolute value at most one, and the suffix vector has norm at most one, so the squared change in ψ is at most the squared change in z_j . Replace the stage vectors one at a time and apply Cauchy–Schwarz to obtain

$$\|\psi(v) - \psi(w)\|^2 \leq N \sum_{j=1}^N \|z_j(v) - z_j(w)\|^2 \leq N\kappa^2 \sum_{j=1}^N \|U_j(v) - U_j(w)\|^2. \quad (6.4)$$

Omitting the final remainder cutoff can only decrease the left side. Here κ is a common upper bound for the stage Lipschitz constants, with $\kappa \leq C[1 + \log(d+1)]^{7/2}$ by (5.4).

Apply the multiple-envelope bound of Lemma 4.2 to (6.4). For every $x \in \mathbb{R}^d$, it gives

$$\sum_{j,i} \mathcal{E}(\psi_{j,i} Fx) \leq (1 + 80N^2\kappa^2)\nu\|x\|^2. \quad (6.5)$$

Changing coefficient projections. Fix $x \in \mathbb{R}^d$. For each j , apply (6.5) to the coefficient vector $\Pi_{j-1}x$ and keep only the nonnegative energy terms from stage j . Since $X_{j,i}x = \psi_{j,i} F \Pi_{j-1}x$, summing the resulting bounds gives

$$\sum_{j,i} \mathcal{E}(X_{j,i}x) \leq (1 + 80N^2\kappa^2)\nu \sum_{j=1}^N \|\Pi_{j-1}x\|^2 \leq N(1 + 80N^2\kappa^2)\nu\|x\|^2. \quad (6.6)$$

This estimate does not require the coefficient projections to commute with Λ . Since $N = O(1 + \log(d+1))$ and $\kappa^2 = O([1 + \log(d+1)]^7)$, the factor $N(1 + 80N^2\kappa^2)$ is at most $C[1 + \log(d+1)]^{10}$. This proves the required matrix energy bound; together with (6.3) and the covariance-tail bound, it completes the proof. $\square$

7 Exactly the required number of cuts

The last step uses the trace-tail information to cap the number of local modes needed from each support at two.

Lemma 7.1. *Suppose finitely many fields $X_i : \mathbb{R}^d \rightarrow L^2(\pi)$ have pairwise disjoint nonempty supports B_i and satisfy*

$$\sum_i X_i^* X_i \succeq \gamma I_d, \quad \sum_i X_i^* \Delta X_i \preceq \mathcal{T} I_d, \quad \text{tail}_1(X_i^* X_i) \leq \delta < \gamma,$$

where $\mathcal{T} > 0$ and $\delta \geq 0$. Set $\lambda_{\text{cut}} = 2\mathcal{T}/(\gamma - \delta)$, and let n_i be the number of eigenvalues of $\Delta_{B_i}^D$ at most λ_{cut} , counted with multiplicity. Then

$$\sum_i \min(n_i, 2) \geq d. \quad (7.1)$$

Proof. Suppose the sum in (7.1) is $n_{\text{cap}} < d$, and let n_{two} be the number of domains with $n_i \geq 2$. For each such domain, choose a unit eigenvector u_i for the largest eigenvalue of $X_i^* X_i$ and impose the coefficient constraint $x \perp u_i$ on $x \in \mathbb{R}^d$. For each domain with $n_i \leq 1$, choose an orthonormal basis of the spectral subspace of Δ_B^D for eigenvalues at most λ_{cut} , and impose $x \perp X_i^* \varphi$ for each basis vector φ . This imposes at most $n_i \leq 1$ constraints. The total number of constraints is at most $n_{\text{two}} + (n_{\text{cap}} - 2n_{\text{two}}) = n_{\text{cap}} - n_{\text{two}}$. Their common kernel contains $d_0 = d - n_{\text{cap}} + n_{\text{two}} > n_{\text{two}}$ orthonormal coefficient vectors $x_1, \dots, x_{d_0} \in \mathbb{R}^d$.

Applying the covariance and energy assumptions to these vectors and summing gives

$$\sum_{i,a} \|X_i x_a\|_\pi^2 \geq \gamma d_0 \quad \text{and} \quad \sum_{i,a} \mathcal{E}(X_i x_a) \leq \mathcal{T} d_0.$$

For a domain with $n_i \geq 2$, the coefficient constraints give $\sum_{a=1}^{d_0} x_a x_a^* \preceq I - u_i u_i^*$. Its total mass is therefore

$$\sum_{a=1}^{d_0} \|X_i x_a\|_\pi^2 = \text{tr} \left(X_i^* X_i \sum_{a=1}^{d_0} x_a x_a^* \right) \leq \text{tr} [X_i^* X_i (I - u_i u_i^*)] = \text{tail}_1(X_i^* X_i) \leq \delta.$$

On every domain with $n_i \leq 1$, each $X_i x_a$ is orthogonal in $L^2(\pi|_{B_i})$ to the spectral subspace for eigenvalues at most λ_{cut} . Expansion in an orthonormal Dirichlet eigenbasis gives $\lambda_{\text{cut}} \|X_i x_a\|_\pi^2 \leq \mathcal{E}(X_i x_a)$. Summing over both types of domains gives

$$\gamma d_0 \leq n_{\text{two}} \delta + \frac{\mathcal{T} d_0}{\lambda_{\text{cut}}} \leq d_0 \delta + \frac{d_0(\gamma - \delta)}{2} = \frac{d_0(\gamma + \delta)}{2} < \gamma d_0,$$

a contradiction. $\square$

Lemma 7.2. *Let $\lambda_{\text{cut}} \geq 0$. If Δ_B^D has at least one eigenvalue at most λ_{cut} , then B supports a nonzero nonnegative function of Rayleigh quotient at most λ_{cut} . If it has at least two eigenvalues at most λ_{cut} , then B supports two such functions with disjoint supports.*

Proof. Replacing a minimizing function by its absolute value cannot increase any original edge energy, so a ground eigenfunction can be chosen nonnegative. This proves the first assertion.

Decompose B into connected components using positive-weight edges with both endpoints in B . The Dirichlet operator is the direct sum of the operators on these components. If two components have ground eigenvalues at most λ_{cut} , take one ground eigenfunction from each. Otherwise one component has two eigenvalues at most λ_{cut} . It remains to prove the assertion on a connected domain, which we again denote by B .

A nonnegative ground eigenfunction f_0 on this connected domain is strictly positive. Indeed, at a vertex with $f_0(v) = 0$, its eigen-equation gives

$$0 = - \sum_{w \in B} P(v, w) f_0(w).$$

Nonnegativity forces f_0 to vanish at every neighbor in B joined to v by an edge of positive weight. Connectedness then gives $f_0 \equiv 0$, a contradiction. Choose an eigenfunction f for the second eigenvalue λ_{loc} of Δ_B^D , orthogonal to f_0 in $L^2(\pi|_B)$. Strict positivity of f_0 forces f to have both signs. Its positive and negative parts, $f_+ = \max\{f, 0\}$ and $f_- = \max\{-f, 0\}$, are consequently nonzero and have disjoint supports. Taking the inner product of $\Delta_B^D(f_+ - f_-) = \lambda_{\text{loc}}(f_+ - f_-)$ with f_+ gives

$$\mathcal{E}(f_+) - \langle f_+, \Delta_B^D f_- \rangle_\pi = \lambda_{\text{loc}} \|f_+\|_\pi^2,$$

because $\langle f_+, f_- \rangle_\pi = 0$. The Dirichlet matrix has nonpositive off-diagonal entries, and hence

$$\mathcal{E}(f_+) = \lambda_{\text{loc}} \|f_+\|_\pi^2 + \langle f_+, \Delta_B^D f_- \rangle_\pi \leq \lambda_{\text{loc}} \|f_+\|_\pi^2.$$

The same argument applied to $-f$ bounds $\mathcal{E}(f_-)$. Both Rayleigh quotients are at most $\lambda_{\text{loc}} \leq \lambda_{\text{cut}}$. All these energies are computed by zero extension in the original graph. $\square$

Lemma 7.3 (Scalar sweep). *Every nonzero nonnegative function f has a nonempty set $S \subseteq \text{supp } f$ with*

$$\phi_G(S) \leq \sqrt{2\mathcal{R}_G(f)}.$$

Proof. For $S_t = \{v : f(v)^2 > t\}$, integrating the contribution of each vertex gives

$$\int_0^\infty \pi(S_t) dt = \|f\|_\pi^2.$$

An edge crosses S_t precisely when t lies between its endpoint values of f^2 , apart from endpoints of the interval. Hence

$$\int_0^\infty c(S_t, S_t^c) dt = \frac{1}{2} \sum_{v,w} c_{vw} |f(v)^2 - f(w)^2|.$$

By Cauchy–Schwarz applied to the edge sum and $(f(v) + f(w))^2 \leq 2(f(v)^2 + f(w)^2)$,

$$\frac{1}{2} \sum_{v,w} c_{vw} |f(v)^2 - f(w)^2| \leq \left[\mathcal{E}(f) \frac{1}{2} \sum_{v,w} c_{vw} (f(v) + f(w))^2 \right]^{1/2} \leq \sqrt{2\mathcal{E}(f)} \|f\|_\pi.$$

The ratio of the two integrals is a volume-weighted average of the conductances of the nonempty level sets. At least one such set has conductance at most the asserted value. $\square$

Proof of Theorem 1.1. If $\lambda_k = 0$, there are at least k connected components. Their indicators give k nonnegative functions with disjoint supports and zero energy, and the components themselves have zero conductance. This also covers $k = 1$.

Otherwise put $d = k$ and $\nu = \lambda_k$. Apply Proposition 6.1 and discard zero fields. Take $\gamma = 5/8$ and the positive energy bound $\mathcal{T} \leq C[1 + \log(k+1)]^{10} \lambda_k$ from that proposition. Since $\delta < 1/4$, we have $\gamma - \delta > 3/8$. Let n_i count the Dirichlet eigenvalues on the support of X_i that are at most

$$\lambda_{\text{cut}} = \frac{2\mathcal{T}}{\gamma - \delta} \leq C[1 + \log(k+1)]^{10} \lambda_k.$$

On that support, Lemma 7.2 gives $\min(n_i, 2)$ nonzero nonnegative functions with disjoint supports and Rayleigh quotients at most λ_{cut} . The supports of the different fields are disjoint, so all these functions have pairwise disjoint supports. There are at least k of them by Lemma 7.1. Choosing any k proves (1.4). Apply Lemma 7.3 separately to each witness. Each resulting set is contained in its witness support, so all k sets are disjoint, and

$$\max_i \phi_G(S_i) \leq \sqrt{2\lambda_{\text{cut}}} \leq C[1 + \log(k+1)]^5 \sqrt{\lambda_k}.$$

After enlarging the absolute constant to cover both conclusions, this is (1.3). $\square$

Acknowledgments

The author thanks Lap Chi Lau for bringing Theorem 7.17 of his book manuscript to the author’s attention after the first version of this paper was posted.

Disclosure of AI assistance

OpenAI’s ChatGPT and Codex (GPT-6) were used to develop the proof and prepare the manuscript. The human author takes full responsibility for the mathematical arguments and the final manuscript.

References

- [1] Afonso S. Bandeira. *Ten Lectures and Forty-Two Open Problems in the Mathematics of Data Science*. Lecture notes, October 10, 2016, Open Problem 3.3. <https://people.math.ethz.ch/~abandeira/TenLecturesFortyTwoProblems.pdf>.
- [2] Lap Chi Lau. *Algorithmic Spectral Graph Theory*. Book manuscript, September 16, 2026, Section 7.7, Theorem 7.17. <https://cs.uwaterloo.ca/~lapchi/papers/ASGT.pdf>.
- [3] James R. Lee, Shayan Oveis Gharan, and Luca Trevisan. Multiway spectral partitioning and higher-order Cheeger inequalities. *Journal of the ACM*, 61(6), Article 37, 1–30, 2014. arXiv:1111.1055.
- [4] Anand Louis, Prasad Raghavendra, Prasad Tetali, and Santosh Vempala. Many sparse cuts via higher eigenvalues. In *Proceedings of the Forty-Fourth Annual ACM Symposium on Theory of Computing (STOC)*, 2012. doi:10.1145/2213977.2214079. arXiv:1111.0965.
- [5] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012. doi:10.1007/s10208-011-9099-z. arXiv:1004.4389.